

Reduced volume and increased training intensity elevate muscle $\text{Na}^+\text{-K}^+$ pump α_2 -subunit expression as well as short- and long-term work capacity in humans

Jens Bangsbo, Thomas P. Gunnarsson, Jesper Wendell, Lars Nybo and Martin Thomassen

J Appl Physiol 107:1771-1780, 2009. First published 1 October 2009;
doi:10.1152/jappphysiol.00358.2009

You might find this additional info useful...

This article cites 60 articles, 27 of which can be accessed free at:

</content/107/6/1771.full.html#ref-list-1>

This article has been cited by 10 other HighWire hosted articles, the first 5 are:

The effects of knee injury on skeletal muscle function, Na^+ , K^+ -ATPase content, and isoform abundance

Ben D. Perry, Pazit Levinger, Hayden G. Morris, Aaron C. Petersen, Andrew P. Garnham, Itamar Levinger and Michael J. McKenna

PHY2, February 1, 2015; 3 (2): .

[\[Abstract\]](#) [\[Full Text\]](#) [\[PDF\]](#)

Concurrent speed endurance and resistance training improves performance, running economy, and muscle NHE1 in moderately trained runners

Casper Skovgaard, Peter M. Christensen, Sonni Larsen, Thomas Rostgaard Andersen, Martin Thomassen and Jens Bangsbo

J Appl Physiol, November 15, 2014; 117 (10): 1097-1109.

[\[Abstract\]](#) [\[Full Text\]](#) [\[PDF\]](#)

The effects of osteoarthritis and age on skeletal muscle strength, $\text{Na}^+\text{-K}^+$ -ATPase content, gene and isoform expression

Ben D. Perry, Pazit Levinger, Fabio R. Serpiello, Marissa K. Caldow, David Cameron-Smith, John R. Bartlett, Julian A. Feller, Neil R. Bergman, Itamar Levinger and Michael J. McKenna

J Appl Physiol, November 15, 2013; 115 (10): 1443-1449.

[\[Abstract\]](#) [\[Full Text\]](#) [\[PDF\]](#)

Upregulation of heme oxygenase-1 potentiates EDH-type relaxations in the mesenteric artery of the spontaneously hypertensive rat

Zhuoming Li, Yu Wang, Ricky Y. K. Man and Paul M. Vanhoutte

Am J Physiol Heart Circ Physiol, November 15, 2013; 305 (10): H1471-H1483.

[\[Abstract\]](#) [\[Full Text\]](#) [\[PDF\]](#)

Quantification of Na^+ , K^+ pumps and their transport rate in skeletal muscle: Functional significance

Torben Clausen

J Gen Physiol, October , 2013; 142 (4): 327-345.

[\[Abstract\]](#) [\[Full Text\]](#) [\[PDF\]](#)

Updated information and services including high resolution figures, can be found at:

</content/107/6/1771.full.html>

Additional material and information about *Journal of Applied Physiology* can be found at:

<http://www.the-aps.org/publications/jappl>

This information is current as of March 11, 2015.

Reduced volume and increased training intensity elevate muscle $\text{Na}^+\text{-K}^+$ pump α_2 -subunit expression as well as short- and long-term work capacity in humans

Jens Bangsbo, Thomas P. Gunnarsson, Jesper Wendell, Lars Nybo, and Martin Thomassen

Copenhagen Muscle Research Centre, Department of Exercise and Sport Sciences, Section of Human Physiology, University of Copenhagen, Copenhagen, Denmark

Submitted 12 April 2009; accepted in final form 28 September 2009

Bangsbo J, Gunnarsson TP, Wendell J, Nybo L, Thomassen M. Reduced volume and increased training intensity elevate muscle $\text{Na}^+\text{-K}^+$ pump α_2 -subunit expression as well as short- and long-term work capacity in humans. *J Appl Physiol* 107: 1771–1780, 2009. First published October 1, 2009; doi:10.1152/jappphysiol.00358.2009.—The present study examined muscle adaptations and alterations in work capacity in endurance-trained runners as a result of a reduced amount of training combined with speed endurance training. For a 6- to 9-wk period, 17 runners were assigned to either a speed endurance group with a 25% reduction in the amount of training but including speed endurance training consisting of six to twelve 30-s sprint runs 3–4 times/wk (SET group $n = 12$) or a control group ($n = 5$), which continued the endurance training (~ 55 km/wk). For the SET group, the expression of the muscle $\text{Na}^+\text{-K}^+$ pump α_2 -subunit was 68% higher ($P < 0.05$) and the plasma K^+ level was reduced ($P < 0.05$) during repeated intense running after 9 wk. Performance in a 30-s sprint test and the first of the supramaximal exhaustive runs was improved ($P < 0.05$) by 7% and 36%, respectively, after the speed endurance training period. In the SET group, maximal O_2 uptake was unaltered, but the 3-km (3,000-m) time was reduced ($P < 0.05$) from 10.4 ± 0.1 to 10.1 ± 0.1 min and the 10-km (10,000-m) time was improved from 37.3 ± 0.4 to 36.3 ± 0.4 min (means $\pm$ SE). Muscle protein expression and performance remained unaltered in the control group. The present data suggest that both short- and long-term exercise performances can be improved with a reduction in training volume if speed endurance training is performed and that the $\text{Na}^+\text{-K}^+$ pump plays a role in the control of K^+ homeostasis and in the development of fatigue during repeated high-intensity exercise.

fatigue; running economy; performance; potassium; $\text{Na}^+\text{-K}^+\text{-Cl}^-$ cotransporter isoform 1

WORK CAPACITY at different exercise intensities is determined by various factors. One way to study the importance of such factors is to change their regulatory systems by performing exercise training and then to examine the physiological response and work capacity during various types of exercise.

Muscle ion transport proteins involved in the exchange of H^+ , Na^+ , K^+ , Cl^- , and lactate across the sarcolemma appear to be of importance in delaying fatigue during intense exercise (10, 47, 48). The $\text{Na}^+\text{-K}^+$ pump is pivotal in maintaining the muscle membrane potential during exercise (11), and $\text{Na}^+\text{-K}^+\text{-Cl}^-$ cotransporter isoform 1 (NKCC1) protein, primarily located in the sarcolemma, may also contribute to the maintenance of muscle function during intense exercise, possibly by adding to K^+ reuptake (62). In untrained human subjects, as

first described in rat muscles (35), the $\text{Na}^+\text{-K}^+$ pump has been shown to be upregulated by different types of exercise training, detected either as content of functional pumps by [^3H]ouabain binding (25, 36, 41) or subunit specific by Western blot analysis (12, 45, 47). In addition, Nielsen et al. (47) observed that elevated levels of $\text{Na}^+\text{-K}^+$ pump α_1 - and α_2 -subunits after 8 wk of knee extensor training at supramaximal exercise intensities (speed endurance training) were associated with a reduced muscle interstitial K^+ concentration during exercise as well as better performance during intense exercise (47). Likewise, Iaia et al. (31) observed that well-trained subjects after a change from endurance to sprint training for 4 wk had a 29% higher expression of muscle $\text{Na}^+\text{-K}^+$ pump α_1 -subunits. This increase in pump content was associated with a reduced plasma K^+ level during exercise ($P < 0.05$) and improved performance during repeated intense running (31). In accordance, studies on well-trained subjects, who either performed strength training (42) or increased their training intensity (17, 40), have reported increased $\text{Na}^+\text{-K}^+$ pump concentrations as determined by the [^3H]ouabain-binding technique. In contrast, Aughey et al. (2) did not find changes in the abundance of any of the $\text{Na}^+\text{-K}^+$ pump α - and β -isoforms when already trained subjects performed a period of intensified training. The lack of effect in the latter study may have been a result of the exercise intensity being below the one corresponding to maximal O_2 uptake ($\dot{V}\text{O}_{2\text{max}}$). The effect of training on NKCC1 has been investigated in rats (22) and in one human study (31), where a tendency to a higher amount of NKCC1 ($\sim 14\%$) was observed for endurance-trained subjects after a period with sprint training. Thus, there is a further need to study the adaptations of these transport systems in trained subjects and a possible importance for work capacity during intense exercise.

Proteins controlling H^+ efflux from the muscle cell may be of importance for the work capacity of a contracting muscle (3, 18). Na^+/H^+ exchanger isoform 1 (NHE1) has been reported to increase as a result of high-intensity exercise training in rats and humans (33) and even when endurance-trained subjects changed to sprint training (31), which was associated with improved short-term performance. In human skeletal muscle, monocarboxylate transporters 1 and 4 (MCT1 and MCT4) facilitate lactate and H^+ exchange across the muscle membrane (32). A number of studies, including either endurance or high-intensity training programs of untrained subjects, have observed higher MCT1 protein density, and some have also observed higher MCT4 protein density (8, 33, 45). On the other hand, in a recent study (31) with endurance-trained subjects, neither the expression of MCT1 nor MCT4 increased after a sprint training period. This may be due to a reduced amount of

Address for reprint requests and other correspondence: J. Bangsbo, Dept. of Exercise and Sport Sciences, August Krogh Bldg., Universitetsparken 13, Copenhagen DK-2100, Denmark (e-mail: jbangsbo@aki.ku.dk).

training, since one study (5) has reported sprint training-induced changes in MCT1 transport proteins in endurance-trained subjects when the subjects maintained a high volume of training (~ 50 km/wk). Thus, the findings may suggest that, for trained athletes, a basic volume of training including frequent sessions of high-intensity exercise are necessary to change muscle MCT1 protein content and lactate transport capacity, but this needs to be studied.

Performance in distance running is a function of $\dot{V}O_{2\max}$, relative work intensity, and running economy, which is defined as the energy used at a given submaximal running speed (13). In endurance-trained subjects, no or small changes in $\dot{V}O_{2\max}$ are observed after a period with more intense training, whereas running economy has been shown to increase after a period of interval (6, 21, 27, 56), plyometric (50, 54, 57, 60), and strength (44) training. In a number of studies (50, 57, 58), better running economy has been associated with performance improvements. On the other hand, in a recent study (30), energy expenditure was diminished by 5–8% after a period with speed endurance training without a change in 10-km (10,000-m) performance, which may be due to a marked reduction in the total amount of training. Therefore, it would be of great interest to examine whether performance during long-term exercise and related variables can change by performing speed endurance training with a moderate reduction in the volume of training.

Therefore, the aim of the present study was to examine the effect of a reduced amount of training combined with speed endurance training on the adaptations of skeletal muscle ion transport proteins and their relation to the physiological response to exercise as well as short- and long-term work capacity. Seventeen endurance-trained subjects were studied before and after a 6- or 9-wk period where five subjects followed their normal training and twelve subjects reduced the volume of training and added sessions of speed endurance training.

METHODS

Subjects

Seventeen moderately trained male endurance runners took part in the study. All subjects were healthy nonsmokers, and none were on medication. Age, height, weight, and $\dot{V}O_{2\max}$ were 34.8 ± 1.5 yr, 182.8 ± 1.5 cm, 74.0 ± 2.0 kg, and 63.0 ± 1.9 ml·kg⁻¹·min⁻¹, respectively (means $\pm$ SE). Subjects had been training and competing on a regular basis for a minimum of 5 yr, and before the study they were all running 4–5 days/wk with an average weekly distance of ~ 55 km. All participants were fully informed of any possible risks and discomforts associated with the experimental procedures before they gave their written informed consent to participate. This study conformed with the code of Ethics of the World Medical Association (Declaration of Helsinki) and was approved by the Ethics Committee of Copenhagen and Frederiksberg communities.

Intervention Period and Training

An intervention (IT) period lasting for 6–9 wk was carried out in the last period of the competitive season. A parallel two-group, longitudinal (pre, post) design was used. Subjects were matched based on their seasonal best running performances (3,000 and 10,000 m) and randomly assigned to either an experimental speed endurance training group (SET group; $n = 12$) or a control group ($n = 5$). Four of the subjects in the SET group carried out a 6-wk IT period, whereas the

remaining eight subjects conducted a 9-wk IT period. During the 9-wk IT period, the control group continued the training as during the period before the study (Fig. 1). During the IT period, the SET group carried out two to three sessions of speed endurance training a week. Each session consisted of 8–12 repeated 30-s running bouts at $\sim 95\%$ of maximal speed separated by 3 min of passive recovery. In addition, about once a week, the SET group did high-intensity aerobic training consisting of 4×4 -min running at an intensity resulting in a heart rate (HR) $>85\%$ of maximal HR ($HR_{\max}$; 1,100–1,250 m) separated by 2 min of passive recovery and one to two sessions of aerobic low-intensity ($HR <75\%$ of $HR_{\max}$) or moderate-intensity (75 – 85% of $HR_{\max}$) training. The total distance covered per week for the control and SET groups was 51.5 ± 3.6 and 33.2 ± 1.6 km, respectively (Fig. 1). The training speed was checked on a regular basis and modified

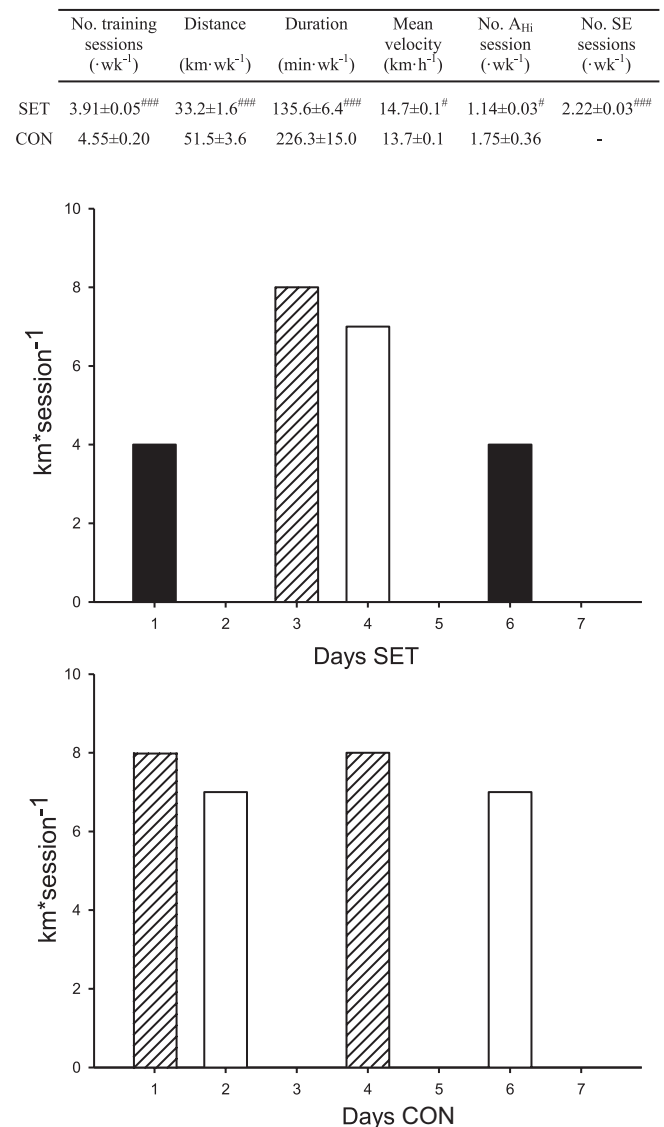


Fig. 1. Overview of the training performed during the intervention period for the control (CON) and speed endurance trained (SET) groups. *Top*: data for the total number of training sessions, distance covered, duration of the training session (expressed per week), mean velocity, number of high-intensity aerobic (A_{HI}) training sessions, and speed endurance (SE) training sessions are presented as means $\pm$ SE. $\#P < 0.05$ and $###P < 0.001$, significant difference between the control and SET groups. *Bottom*: 1 wk of training for the SET and control groups with speed endurance training (black bars), high-intensity aerobic training (open bars), and aerobic training (hatched bars) sessions.

every 14 days. In the SET group, all high-intensity aerobic and speed endurance training sessions were performed on a track and were carefully supervised. At the beginning of each training session, subjects were running 4.3 km with an average speed of ~ 13 km/h, and after the speed endurance training they ran ~ 1 km at a speed of ~ 11 km/h to recover. Subjects maintained their habitual life style and normal daily food intake. HR was recorded (Polar S610 HR monitor, Polar Electro, Kempele, Finland) in all training sessions in the IT period and immediately downloaded on portable personal computer after each training session. In addition, all training sessions for the control group were recorded and analyzed every week.

Experimental Protocol

Before and after the IT period, the participants completed the following tests: 1) a 30-s sprint test (only the SET group); 2) four 6-min runs at different submaximal running velocities (12, 14, 16, and 17 km/h) interspersed by 2 min of rest (SUB test) after a 10-min pause followed an incremental treadmill test (INC test) to exhaustion to determine $\dot{V}O_{2\max}$; 3) two exhaustive supramaximal treadmill runs (EX1 and EX2) at a speed of $\sim 130\%$ $\dot{V}O_{2\max}$ separated by 2 min of passive rest [repeated supramaximal sprint (RS) test]; 4) a 3,000-m race on a track (3-K); and 5) a 10,000-m race on a track (10-K). Before the experiment, all subjects had high experience with treadmill running and 3- and 10-km racing on a track. The INC, EX1, EX2, and 30-s sprint tests were preceded by pretests to familiarize the subjects with the testing procedures. A muscle biopsy was taken before testing 48 h after the last training session.

All sets of tests were carried out at least 2 days apart, and the order of the tests was the same for the control and SET groups. The 3-km test was performed 48 h before the last speed endurance training session, and the 30-s sprint test was performed at the start of the last session. Forty-eight hours after the last training session, a muscle biopsy was obtained at rest, and the INC test was carried out after. The RS test was performed after another 48 h, and, after 24 h, subjects then underwent another speed endurance training session. After another 48 h of recovery, subjects completed the 10-km test. The control and SET groups performed the same number of training session between the tests.

Testing Procedures

On the day of testing, subjects reported to the laboratory 3 h after consuming a light meal. Subjects refrained from strenuous physical activity in the 48 h before testing and abstained from alcohol and caffeine consumption 24 h before testing. To minimize the effect of diet on muscle metabolism and performance, 2 days before any experimental testing the participants were also required to follow a nutritional strategy designed to ensure an adequate carbohydrate intake ($\sim 60\%$ of total energy intake) and to record and replicate their individual dietary pattern during the 48 h before each testing day. All tests were preceded by 15 min of standardized warm-up.

A 30-s sprint test was carried out on a track, and the distance was recorded (only the SET group). Subjects completed the SUB, INC, and RS tests on a motorized treadmill under standard laboratory conditions. The calibration of the treadmill was checked before each testing session. Before the tests, subjects had a Polar S610 HR monitor (Polar Electro) fitted around their chests for continuous HR recordings, and a catheter (18 gauge, 32 mm) was inserted in an antecubital vein (not the subjects with a 6-wk IT period). The INC test started with a 3-min run at a preset speed (14 km/h), after which the speed was increased by 1 km/h every minute until volitional fatigue. Pulmonary $\dot{V}O_2$ was measured throughout the whole protocol by a breath-by-breath gas analyzing system (Jaeger MasterScreen, CPX, Viasys Heathcare, Hoechberg, Germany). The analyzer was automatically calibrated before each test with a gas of known O_2 and CO_2 concentration (53). $\dot{V}O_{2\max}$ was determined as the highest value achieved over a 30-s period. A plateau in $\dot{V}O_2$ despite an increased

power output and a respiratory exchange ratio (RER) of >1.15 were used as criteria for $\dot{V}O_{2\max}$ achievement. Blood samples were taken at rest and at the end of each exercise bout as well as 3, 6, and 9 min after the exhaustive run. All blood samples were collected in 2-ml heparinized syringes.

In the RS test, subjects performed two supramaximal exhaustive runs (EX1 and EX2) at a speed corresponding to $\sim 130\%$ pretraining $\dot{V}O_{2\max}$ separated by a 2-min rest period. The speed was determined by a linear extrapolation established from the individual relationship between exercise intensity and pulmonary $\dot{V}O_2$ obtained during the INC test. After a 10-min warm-up (at a running speed of 14 km/h), subjects rested for 5 min before starting the EX1 test. The exercises were terminated when the subject failed to maintain the speed. Subjects were not given any feedback. Pulmonary $\dot{V}O_2$ was measured as previously described. HR was collected at 5-s intervals. Blood samples were taken before and at the end of each exercise bout as well as 1 min after the EX1 test and 1.5 and 3 min after the EX2 test. All blood samples were collected in 2-ml heparinized syringes.

The 3- and 10-km trials were carried out on a 400-m track. To avoid racing tactics and strategies, the test was conducted on an individual basis with participants starting at 1-min intervals in random order. HR was measured before and during the tests with a Polar S610 HR monitor (Polar Electro) fitted around the chest.

Muscle biopsies were taken before and after the IT period (4). A small incision through the skin and fascia over the vastus lateralis muscle was made under local anesthesia (1 ml; 20 mg/l lidocaine without adrenaline). The subject's left or right leg was randomly selected. The same leg was used for pre- and post-IT period biopsies. Tissue samples were immediately frozen in liquid nitrogen and subsequently stored at -80°C .

Blood Analysis

Immediately after being sampled, a part of the blood was rapidly centrifuged at 20,000 g for 30 s. Thereafter, the plasma was transferred to Eppendorf tubes and placed in ice-cold water until being stored at -20°C . Samples were subsequently analysed for K^+ by an ion-selective electrode using a Hitachi 912 Automatic Analyzer (Roche Diagnostic). Another part of the blood sample (100 μl) was hemolyzed using a 1:1 dilution with a buffer solution (Yellow Spring Instruments, Yellow Springs, OH) to which 20 g/l Triton X-100 was added (20) for the analysis of lactate (model 23, Yellow Spring Instruments).

Muscle Analysis

The frozen muscle biopsies were weighed before and after freeze drying to determine the water content. After the freeze drying, all connective tissue, visible fat, and blood were carefully dissected away under a stereo microscope in a room with a temperature of 18°C and a relative humidity below 30%.

Muscle ion transport proteins. Muscle tissue taken at rest (~ 4 – 5 mg dry wt) was homogenized on ice in a fresh batch of buffer (10% glycerol, 20 mM Na-pyrophosphate, 150 mM NaCl, 50 mM HEPES, 1% Nonidet P-40, 20 mM β -glycerophosphate, 10 mM NaF, 2 mM PMSF, 1 mM each EDTA and EGTA and 10 $\mu\text{g}/\text{ml}$ each aprotinin and leupeptin and 3 mM benzamide) with a Polytron 3100 (Kinematica) for not more than 30 s. After being rotated end over end for 1 h at 4°C , samples were centrifuged for 30 min at 17,500 g at 4°C , and lysates were collected as the supernatant. Protein concentrations were determined in the lysates using BSA standards (Pierce).

The lysates were diluted to appropriate protein concentrations in a $\times 6$ sample buffer (0.5 M Tris base, DTT, SDS, glycerol, and bromophenol blue) and then boiled for 3 min at 96°C for protein denaturation. Equal amounts of total protein (5–15 μg in accordance to the antibody optimization) were loaded for each sample in different wells on either a 5% (NKCC1) or 10% precast Tris-HCl gel (Bio-Rad Laboratories). For comparisons, samples from the same subject were

always loaded on the same gel. The gel electrophoresis was done with 55 mA and maximum 150 V per gel in ~80–100 min, and, afterward, proteins were blotted to a polyvinylidene difluoride membrane using 70 mA and maximum 25 V per gel in 2 h. Membranes were incubated overnight with ~10 ml of primary antibody diluted in either 2% nonfat milk [monoclonal Na⁺-K⁺ pump α 1-subunit, 1:500 dilution (α 6F, Iowa Hybridoma Bank and C464.6, no. 05-369, Millipore); monoclonal Na⁺-K⁺ pump α 2-subunit, 1:200 dilution (McB2, kindly donated by K. J. Sweadner to H. Bundgaard); polyclonal α 2-subunit, 1:500 dilution (no. 07-674, Millipore); monoclonal β 1-subunit, 1:1,000 dilution (MA3-930, Affinity BioReagents); and polyclonal NKCC1, 1:200 dilution (Sc-21545, Santa Cruz Biotechnology)] or 3% BSA [monoclonal NHE1, 1:500 dilution; polyclonal MCT1, 1:1,000 dilution; and polyclonal MCT4, 1:1,000 dilution (MAB3140, AB3538P, and AB3316P, all from Millipore)]. After a brief wash in Tris-buffered saline-Tween, membranes were incubated with secondary antibody for 1 h at room temperature. The secondary horseradish peroxidase-conjugated antibodies used were diluted 1:5,000 in 2% nonfat milk or 3% BSA depending on the primary antibody (P-0447, P-0448, and P-0449, DakoCytomation). Membrane staining was visualized by a 5-min incubation with a chemiluminescent horseradish peroxidase substrate (Millipore) immediately before the membrane image was digitalized (KODAK Image Station 2000MM). The net band intensities were quantified as the total intensity minus the background intensity (Molecular Imaging Software, KODAK).

Muscle enzymes. Citrate synthase (CS), β -hydroxyacyl-CoA dehydrogenase (HAD), creatine kinase (CK), and phosphofructokinase (PFK) activity were determined fluorometrically on whole muscle (2 mg dry wt) homogenized (1:400) in 0.3 M phosphate-BSA buffer adjusted to pH 7.7 (39).

Statistics

Before the IT period, Student's unpaired *t*-tests were used to compare subjects' characteristics between the two training groups (SET vs. control group). Two-way ANOVA for repeated measurements with control versus SET as a factor and pre versus post as a factor were used to determine the effects on muscle enzyme activity, relative HR, $\dot{V}O_{2\max}$, and exercise performance (3-km run, 10-km run, and INC test). Separate for the SET and control groups, two-way ANOVA for repeated measurements with pre and post as a factor and the different sample times as a factor were used to determine the effects on blood variables during the SUB, INC, and RS tests, exercise performance in the RS test, running economy, and RER. To determine the effects on 30-s sprint test performance, we used one-way ANOVA for repeated measurements and a paired *t*-test. When an overall statistical difference was obtained, we used the Student-Newman-Keuls method as a multiple-comparison procedure to isolate which groups differed from the others.

Changes in muscle protein expression were examined on log-transformed data by applying a paired *t*-test for the SET and control groups. The individual signal intensity was related to the group pre mean before log transformation. From these log values, individual data points were left out in case of a >2-SD difference to the group mean value. Furthermore, even with continuous analysis, the signal intensities from some of the samples were so weak that a quantitative analysis was impossible; these samples were then left out. Because of these criteria, the following numbers of subjects in the SET group were included in the analysis of the different proteins: *n* = 12, Na⁺-K⁺ pump β 1-subunit and NHE1; *n* = 11, NKCC1 and Na⁺-K⁺ pump α 1- and α 2-subunits; and *n* = 8, MCT1 and MCT4. For all proteins, all five subjects were included in the group. Data obtained with the two different antibodies against the Na⁺-K⁺ pump α 1- and α 2-subunits, respectively, were similar, and data were averaged and expressed only as α 1- and α 2-subunits. For all the analyses, the level of statistical significance was set at *P* < 0.05. Data are presented as

means $\pm$ SE except for muscle protein data, which are presented as geometric mean $\pm$ 95% confidence intervals.

RESULTS

Performance

The SET group improved (*P* < 0.001) performance by 3.3% (pre: 10.4 $\pm$ 0.1 min vs. post: 10.1 $\pm$ 0.1 min) and mean speed at the 3-km run from 17.3 $\pm$ 0.2 to 17.9 $\pm$ 0.2 km/h. 10-km performance was improved by 3.1% (37.3 $\pm$ 0.4 vs. 36.3 $\pm$ 0.4 min) with mean velocity being 16.1 $\pm$ 0.2 km/h before and 16.6 $\pm$ 0.2 km/h after the IT period, whereas the performance of the control group was unaltered (Fig. 2). Six of the runners in the SET group conducted a new personal record on the 10-km after the IT period, with an improvement from 36.7 $\pm$ 0.3 to 35.5 $\pm$ 0.1 min. The time to exhaustion during the INC test was increased (*P* < 0.001) by 9.0% (570 $\pm$ 17 vs. 623 $\pm$ 16 s) for the SET group, resulting in a higher running speed for the SET group (20.5 $\pm$ 0.3 vs. 21.4 $\pm$ 0.3 km/h) compared with the control group (Fig. 3A). The SET group also had a better performance (36%, *P* < 0.001) in the EX1 test (108 $\pm$ 11 vs. 141 $\pm$ 9 s) but not in the EX2 test (70 $\pm$ 5

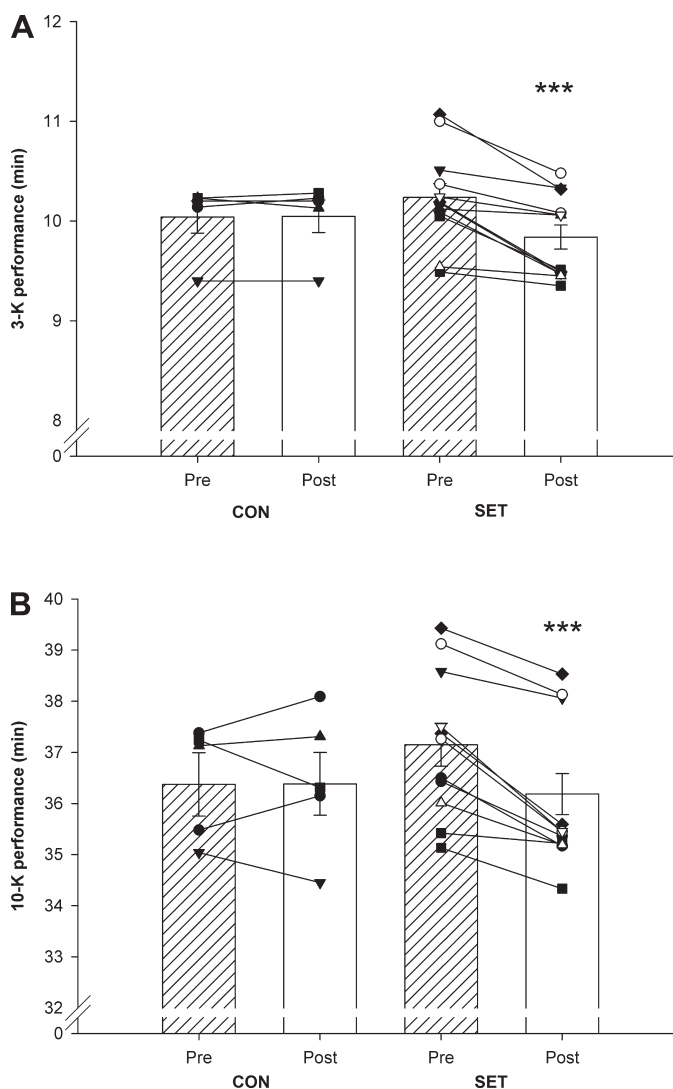


Fig. 2. Mean and individual values of 3-km (A) and 10-km (B) performances before (pre) and after (post) the intervention period in the control (*n* = 5) and SET (*n* = 12) groups divided into subjects covering the 6-wk (open symbols; *n* = 4) and 9-wk (solid symbols; *n* = 8) periods. ****P* < 0.001, significant difference between pre and post values.

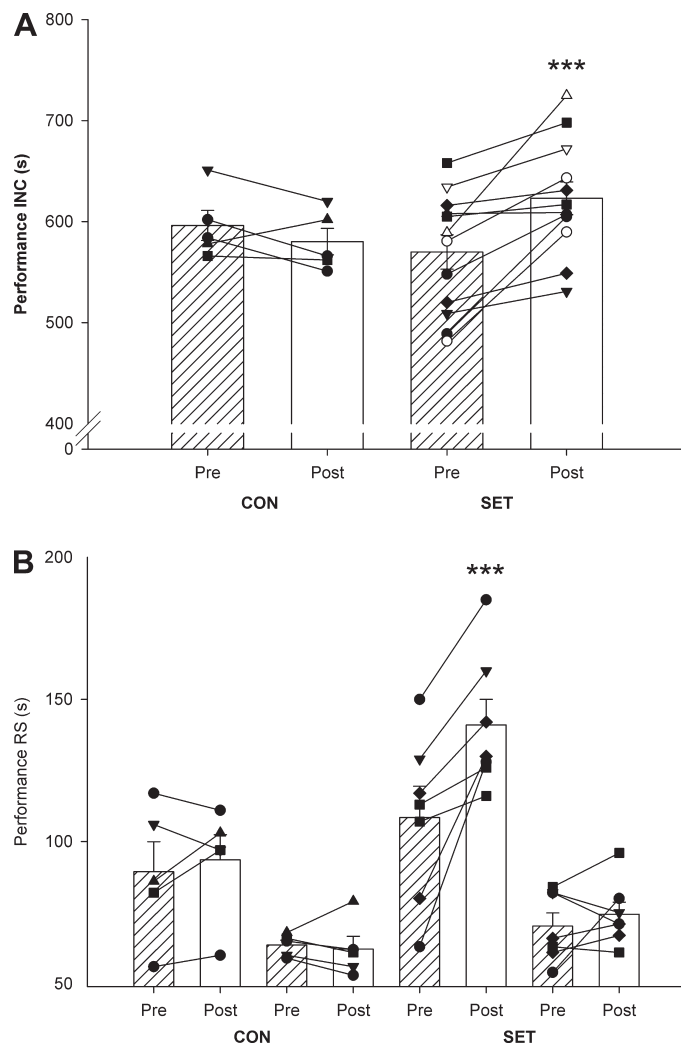


Fig. 3. Mean and individual values of performance during the incremental (INC) tests (A) and in the first (left) and second (right) runs of the repeated supramaximal sprint (RS) test (B) before (pre) and after (post) the intervention period in the control ($n = 5$) and SET ($n = 12$) groups divided into subjects covering the 6-wk (open symbols; $n = 4$) and 9-wk (closed symbols; $n = 8$) periods. *** $P < 0.001$, significant difference between pre and post values.

vs. 74 ± 4 s) and RS test (Fig. 3B). Performance of the control group in the INC and RS tests was the same before and after the IT period (Fig. 3). The runners in the SET group who had a 6-wk training period ($n = 4$) had a 11.0 ± 1.5 -m (4.9%) improvement ($P < 0.05$) in the 30-s maximal running test, whereas the improvement ($P < 0.001$) in performance for the runners training for 9 wk ($n = 8$) was 10.8 ± 1.8 m (5.3%) and 14.0 ± 1.8 m (6.8%) after 6 and 9 wk, respectively. In the last 3 wk, the distance was increased ($P < 0.05$) by 3.3 ± 0.8 m (1.5%).

HR During 3- and 10-km Runs

In the SET group, the mean HR during the 3-km run was 175.5 ± 1.3 and 176.0 ± 2.6 beats/min, corresponding to $92.5 \pm 1.1\%$ and $92.7 \pm 0.9\%$ of HR_{max} , before and after the IT period, respectively. In the control group, the mean HR was 168.0 ± 4.4 ($92.5 \pm 1.1\%$) and 168.4 ± 3.8 ($92.7 \pm 0.9\%$) beats/min before and after the IT period, respectively. During 10-km run, the HR was 176.2 ± 2.6 ($92.4 \pm 0.5\%$) and 176.7 ± 1.5 ($92.5 \pm 0.8\%$) beats/min, respectively, in the SET group and 168.8 ± 2.5 ($93.7 \pm 0.8\%$) and 167.0 ± 2.9 ($92.7 \pm 0.7\%$) beats/min, respectively, in the control group.

$\dot{V}O_{2max}$

There were no changes in $\dot{V}O_{2max}$ for either the SET (4.44 ± 0.19 vs. 4.56 ± 0.19 l O_2 /min) or control (5.14 ± 0.16 vs. 5.04 ± 0.08 l O_2 /min) groups. Correspondingly, $\dot{V}O_{2max}$ expressed per body weight was not changed for either the SET (61.0 ± 2.4 vs. 62.5 ± 1.9 ml $O_2 \cdot min^{-1} \cdot kg^{-1}$) or control (67.8 ± 2.4 vs. 67.0 ± 2.6 ml $O_2 \cdot min^{-1} \cdot kg^{-1}$) groups, and there were no differences between the groups.

Muscle Ion Transport Proteins

The expression of the Na^+K^+ pump α_2 -subunit was 68% higher ($P < 0.05$) after the IT period (Fig. 4B), and the Na^+K^+ pump β_1 -subunit tended to be higher (10%, $P = 0.053$; Fig. 4C), whereas the expression of the Na^+K^+ pump α_1 -subunit was unaltered (Fig. 4A). The expression of MCT1, MCT4, NHE1, and NKCC1 was not changed in the SET group (Table 1). The control group had no differences in the expression of any of the ion transport proteins (Fig. 4, A–C, and Table 1). The activity of HAD, CS, PFK, and CK was not changed during the IT period in either SET or control groups (Table 2).

Physiological Response to Exercise

Running economy. After the IT period, the SET group improved running economy at 12 km/h ($P < 0.05$) from 199 ± 7 to 193 ± 6 ml $O_2 \cdot kg^{-1} \cdot km^{-1}$, whereas at 14, 16, and 17 km/h, no significant differences were reached (Table 3). In the control group, no significant differences were observed (Table 3).

RER values. In the SET group, RER values at 12, 14, 16, and 17 km/h before and after the IT period were 0.933 ± 0.017 vs. 0.932 ± 0.015 , 0.966 ± 0.019 vs. 0.948 ± 0.015 , 1.003 ± 0.019 vs. 0.982 ± 0.015 , and 1.056 ± 0.025 vs. 1.008 ± 0.012 , respectively, with a significantly ($P < 0.01$) lower value at 17 km/h after the IT period (Table 3). In the control group, no significant differences in RER values were observed (Table 3).

Plasma K^+ Concentration

The K^+ concentration at rest, during the SUB test, and after the INC test was the same before and after the IT period in both the SET and CON groups (Table 4). In the SET group, the plasma K^+ concentration at the end of each of the two exhaustive runs (EX1 and EX2 tests) was lower ($P < 0.01$) after the IT period (Table 5), whereas no differences in K^+ concentration observed before and after the RS test. No differences in K^+ concentration during the RS test were observed in the control group (Table 5).

Blood Lactate Concentration

No differences in blood lactate concentration during the SUB test and at exhaustion in the INC test were observed before and after the training period in either the SET or control groups, but after the IT period, blood lactate concentration in the SET group was lower ($P < 0.01$) 3, 6, and 9 min into recovery (Table 4). Before, during, and after the RS test, no differences in blood lactate concentration were observed in either the SET or control groups (Table 5).

DISCUSSION

The major findings of the present study were that inclusion of speed endurance training with a reduction in training volume not only resulted in improved short-term work capacity but also increased 3- and 10-km performance in endurance-trained runners. The improvements were associated with an $\sim 70\%$ higher expression of Na^+K^+ pump α_2 -subunit and lower plasma K^+ concentrations during exhaustive running.

An amazing finding of the present study was that performance of these well-trained runners in the 3- and 10-km trials

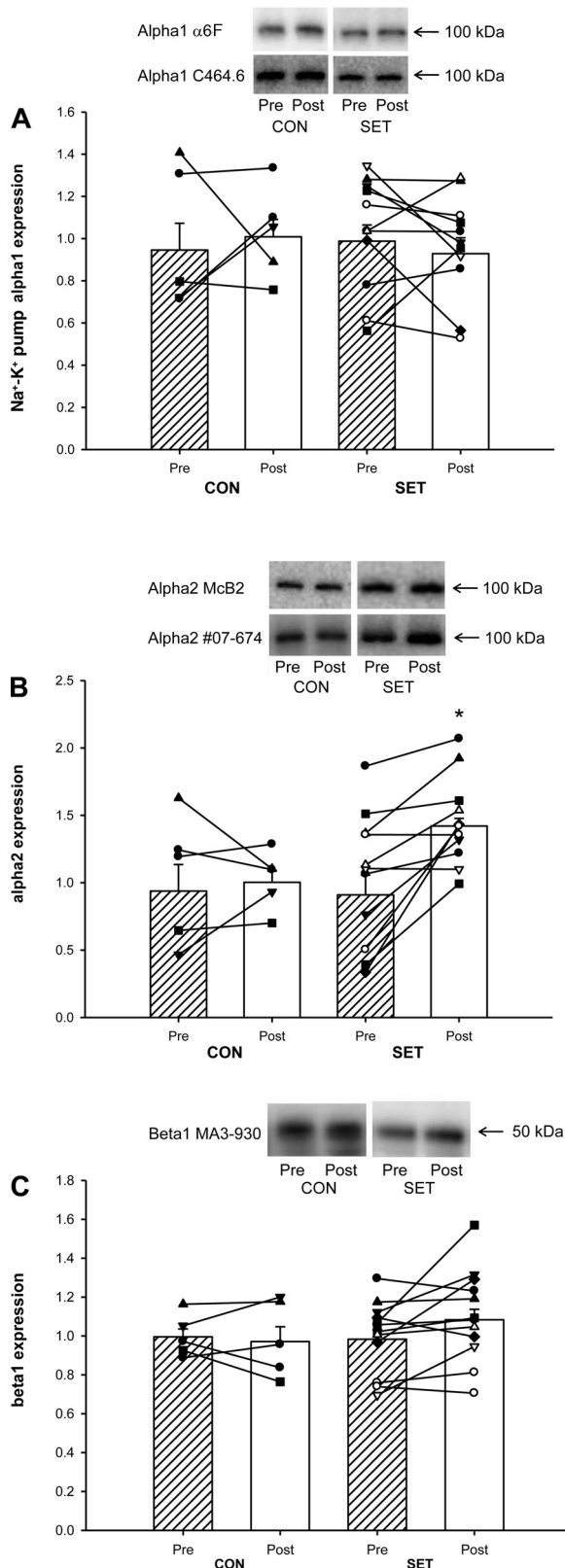


Fig. 4. Bottom: geometric mean values for the protein expression of the $\text{Na}^+\text{-K}^+$ pump α_1 -subunit (A), α_2 -subunit (B), and β_1 -subunit (C) before (pre) and after (post) the intervention period in the control ($n = 5$) and SET ($n = 11\text{--}12$) groups divided into subjects covering the 6-wk (open symbols; $n = 4$) and 9-wk (closed symbols; $n = 7\text{--}8$) periods. Top: representative Western blots for the different antibodies used. * $P < 0.05$, significant difference between pre and post values.

Table 1. Protein expression (post relative to pre) after the intervention period for the SET and control groups

	Control Group		SET Group	
	Geometric means	95% Confidence interval	Geometric means	95% Confidence interval
Monocarboxylate transporter 1	-5	-42 to 58	-2	-8 to 4
Monocarboxylate transporter 4	-14	-23 to -3	-10	-23 to 5
Na^+/H^+ exchanger isoform 1	-3	-39 to 55	1	-23 to 32
$\text{Na}^+\text{-K}^+\text{-Cl}^-$ cotransporter isoform 1	17	3-33	24	-7 to 67

Values are geometric means $\pm$ 95% CI in %; $n = 5$ subjects in the control group and 8-12 subjects in the speed endurance training (SET) group. Pre, before the intervention period; post, after the intervention period.

was significant elevated as a result of the speed endurance training and reduced amount of training. Furthermore, that six of the runners in the SET group at the end of the IT period made their best 10-km time ever with an improvement of >1 min, despite having been running for >5 yr and performing >20 10-km runs. These observations suggest that speed endurance training is a powerful stimulus to improve performance even in an event lasting >30 min despite that the duration of each exercise bout was 30 s. In our previous study (31), which also examined the effect of speed endurance training, no change in 10-km performance was observed, but in that study the amount of training was reduced by $\sim 80\%$. Apparently, maintaining some aerobic sessions with high-intensity running, together with the speed endurance training, played a key role in causing the better performance in the present study. This is the first study to show increased long-term performance in endurance runners with a reduced volume of training and speed endurance training. Burgomaster et al. (9) found an increased cycle endurance capacity from 26 to 51 min in untrained subjects after a 2-wk period with six sessions of four to seven 30-s sprints, which was associated with elevated levels of the CS activity. In the present study, no changes in the activities of CS and HAD were observed, which may be due to the relative high levels before the IT period. On the other hand, a main effect of diminished RER was observed during submaximal running after the IT period in the SET group, with RER being significant lower at 17 km/h, which may suggest that the degree of fat oxidation was elevated at the velocity used during the 3-km test (17-18 km/h) and 10-km test (16-17 km/h). It may have allowed the same rate of muscle glycogen utilization at the higher running velocities after the IT period and may be part of the explanation of the better performance,

Table 2. Enzyme activity before and after the intervention period in the control and SET groups

	Control Group		SET Group	
	Pre	Post	Pre	Post
Citrate synthase	41.7 ± 7.4	42.3 ± 3.3	45.3 ± 4.7	43.7 ± 4.2
β -Hydroxyacyl-CoA-dehydrogenase	22.1 ± 4.3	21.5 ± 2.9	29.9 ± 3.2	26.6 ± 2.1
Creatine kinase	5651 ± 84	6166 ± 360	5986 ± 254	6606 ± 234
Phosphofructokinase	58.1 ± 8.1	71.9 ± 4.5	95.6 ± 11.0	105.6 ± 13.5

Values are means $\pm$ SE (in $\mu\text{mol} \cdot \text{g dry wt}^{-1} \cdot \text{min}^{-1}$); $n = 3$ subjects in the control group and 8 subjects in the SET group.

Table 3. *Running economy ($\dot{V}O_2$) and RER during the submaximal test before and after the intervention period in the control and SET groups*

	Running Speed			
	12 km/h	14 km/h	16 km/h	17 km/h
<i>Control group</i>				
$\dot{V}O_2$, ml $O_2 \cdot kg^{-1} \cdot km^{-1}$				
Pre	221.3 $\pm$ 10.6	214.3 $\pm$ 10.7	215.1 $\pm$ 9.3	212.2 $\pm$ 8.6
Post	209.5 $\pm$ 6.7	205.0 $\pm$ 8.5	205.4 $\pm$ 8.6	207.1 $\pm$ 9.3
RER				
Pre	0.91 $\pm$ 0.01	0.94 $\pm$ 0.01	0.98 $\pm$ 0.01	1.01 $\pm$ 0.01
Post	0.89 $\pm$ 0.02	0.93 $\pm$ 0.02	0.96 $\pm$ 0.02	0.98 $\pm$ 0.02
<i>SET group</i>				
$\dot{V}O_2$, ml $O_2 \cdot kg^{-1} \cdot km^{-1}$				
Pre	199.4 $\pm$ 6.7	193.5 $\pm$ 6.5	194.9 $\pm$ 6.0	198.3 $\pm$ 5.3
Post	193.0 $\pm$ 5.7*	189.7 $\pm$ 5.2	189.7 $\pm$ 5.3	193.3 $\pm$ 4.7
RER				
Pre	0.93 $\pm$ 0.02	0.97 $\pm$ 0.02	1.00 $\pm$ 0.02	1.06 $\pm$ 0.03
Post	0.93 $\pm$ 0.02	0.95 $\pm$ 0.02	0.98 $\pm$ 0.02	1.01 $\pm$ 0.01†

Values are means $\pm$ SE; $n = 5$ subjects in the control group and 12 subjects in the SET group. $\dot{V}O_2$, O_2 uptake; RER, respiratory exchange ratio. * $P < 0.05$ and † $P < 0.01$, significant difference between pre and post values.

even though the role of muscle glycogen for performance during such events is not clear. $\dot{V}O_{2max}$ was not changed and cannot explain the improvements. A third important factor in long-term running performance is running economy (13). In the present study, a main effect of reduced running economy was observed after the IT period in the SET group, and the running economy was significant lower at 12 km/h. Similarly, Paavolainen et al. (50) found an improved running economy after a period where the subjects performed jump and sprint training, and the time allocated training was maintained. In addition, Iaia et al. (30) observed a better running economy when endurance-trained subjects only carried out speed endurance training. In contrast, Mikkola et al. (43) found an unaltered running economy when the volume was not changed but 19% of the training was performed as sprint, jump, and strength training once a week during an 8-wk period. Taken

together, these studies suggest that sprint and frequent jump training with or without a significant reduction in training volume may lead to improvements in running economy for endurance-trained runners.

The amount of the Na^+-K^+ pump α_2 -subunit was higher after the reduced volume of training and speed endurance training, whereas there was no change in the α_1 -subunit and a tendency to elevated levels of the β_1 -subunit. In contrast, Iaia et al. (31) observed that a change in training from regular endurance running to sprint training resulted in increased levels of muscle Na^+-K^+ pump α_1 -isoforms with no change in either α_2 - or β_1 -isoform levels. The difference may be related to the difference in the amount of endurance training, since it has been shown to increase the expression of Na^+-K^+ pump α_2 - and β_1 -isoform proteins (24), which probably are the most abundant subunits in muscle (26, 29, 49). In the study of Iaia et al. (31), the runners only performed speed endurance training, whereas the subjects in the present study carried out a significant amount of aerobic training in addition to speed endurance training. It appears that the combined high-intensity aerobic and speed endurance training in the present study provided a sufficient stimulus for increasing the level of the α_2 -subunit even though the amount of these isoforms may have been high before the study as the subjects were endurance trained. Likewise, in two studies (45, 47) of sedentary people performing repeated high-intensity training, elevated levels of α_2 -subunits were found. In only one of the studies (47), the level of the α_1 -subunit was higher after training. The lack of increase in the α_1 -subunit, which is in contrast to the finding of Iaia et al. (31), may suggest that a significant reduction in training volume is needed to cause a rise in the α_1 -subunit with speed endurance training in endurance-trained subjects. It may also have been an effect of the reduction in relative number of slow twitch fibers (data not shown), since α_1 -subunit expression has been shown to be higher in slow twitch compared with fast twitch fibers in rats (19). One of the explanations of the pronounced Na^+-K^+ pump upregulation in the present study may have been a higher intracellular Na^+ concentration during the training with high intensity compared with the normal submaximal

Table 4. *Plasma K^+ and lactate concentrations during submaximal running and after the incremental test before and after the intervention period in the control and SET groups*

	Rest	Running Speed				Exhaustion	Recovery		
		12 km/h	14 km/h	16 km/h	17 km/h		3 min	6 min	9 min
Control group									
K ⁺ concentration, mmol/l									
Pre	3.68±0.16	4.26±0.12	4.28±0.21	4.66±0.28	4.92±0.17	5.40±0.12	3.44±0.15	3.18±0.22	3.38±0.15
Post	3.74±0.16	4.46±0.30	4.38±0.18	4.66±0.21	4.86±0.17	5.40±0.18	3.62±0.25	3.26±0.19	3.38±0.15
Lactate concentration, mmol/l									
Pre	0.66±0.05	0.94±0.13	1.09±0.15	1.66±0.34	2.78±0.62	7.07±0.95	7.25±1.05	6.55±1.00	5.93±1.07
Post	0.69±0.09	0.93±0.27	0.97±0.23	1.82±0.51	3.07±0.82	5.72±0.33	5.96±0.66	5.27±0.73	4.48±0.88
SET group									
K ⁺ concentration, mmol/l									
Pre	3.91±0.13	4.29±0.06	4.33±0.04	4.73±0.16	4.76±0.05	4.99±0.06	3.44±0.14	3.34±0.05	3.59±0.11
Post	3.70±0.13	4.26±0.10	4.35±0.13	4.49±0.12	4.61±0.08	4.97±0.27	3.66±0.18	3.31±0.09	3.31±0.14
Lactate concentration, mmol/l									
Pre	0.78±0.07	1.00±0.14	1.22±0.20	2.02±0.30	3.65±0.45	7.08±0.73	7.41±0.55	7.62±0.57	6.21±0.30
Post	0.81±0.07	1.06±0.08	1.12±0.09	1.92±0.11	3.24±0.18	8.31±0.71	8.72±0.59*	8.47±0.64*	7.87±0.77†

Values are means $\pm$ SE; $n = 5$ subjects in the control group and 8 subjects in the SET group. * $P < 0.05$ and † $P < 0.01$, significant difference between pre and post values.

Table 5. Plasma K^+ and lactate concentrations during and after the repeated supramaximal sprint test before and after the intervention period in the control and SET groups

	Rest	EX1	Recovery Period 1		EX2	Recovery Period 2		
			1 min	2 min		1.5 min	3 min	6 min
Control group								
K ⁺ concentration, mmol/l								
Pre	3.94±0.10	5.93±0.30	5.06±0.57	4.24±0.21	5.33±0.11	4.34±0.09	3.46±0.11	3.24±0.09
Post	4.28±0.23	5.70±0.12	4.74±0.20	4.32±0.27	5.53±0.03	4.22±0.27	3.56±0.12	3.32±0.09
Lactate concentration, mmol/l								
Pre	0.73±0.06	7.44±1.03	7.30±0.95	7.67±0.89	10.69±1.40	10.29±0.93	10.50±0.89	10.56±0.95
Post	0.77±0.11	7.06±1.22	6.14±0.70	6.62±0.53	9.89±1.30	8.86±1.11	8.95±0.89	8.59±0.89
SET group								
K ⁺ concentration, mmol/l								
Pre	3.89±0.03	5.63±0.21	4.86±0.16	4.44±0.21	5.46±0.22	4.24±0.07	3.45±0.06	3.31±0.06
Post	3.89±0.07	5.30±0.19*	4.71±0.17	4.30±0.14	4.85±0.15†	3.94±0.14	3.41±0.10	3.33±0.11
Lactate concentration, mmol/l								
Pre	0.85±0.07	8.04±0.90	6.85±0.68	7.51±0.72	10.55±1.01	10.01±0.72	10.44±0.80	10.05±0.75
Post	0.92±0.08	9.41±0.75	7.30±0.95	7.67±0.89	10.69±1.40	10.29±0.93	10.50±0.89	10.56±0.95

Values are means ± SE; $n = 5$ subjects in the control group and 8 subjects in the SET group. EX1 and EX2, exhaustive supramaximal treadmill run test 1 and 2, respectively. * $P < 0.01$ and † $P < 0.001$, significant difference between pre and post values.

endurance training (55), since intracellular Na^+ is a possible transcription factor for the Na^+-K^+ pump (46). Thus, Ladka and Ng (37) used the Na^+ channel activator veratridine to modulate intracellular Na^+ and observed increased α_2 -subunit protein expression in C₂C₁₂ skeletal muscle cells. Furthermore, increased intracellular Na^+ was related to an upregulation of Na^+-K^+ ATPase molecules in cultured chick skeletal muscles (61).

The present study showed that performance during the INC test and during the first intense exercise bout to exhaustion (EX1; ~2 min) was elevated despite the reduction in the volume of training, which is in accordance with the study of Iaia et al. (31), where speed endurance training of the same type as in the present study was used and where the amount of training was lowered by 64%. A number of other studies (5, 15, 28, 38, 59) have demonstrated that performance during exercise lasting <5 min can be improved after a period including high-intensity exercise training. However, in contrast to these studies, the amount of training was significantly reduced in the present study and in our previous study (31), which suggests that the intensity, rather than the amount, of training is the main factor in performance improvement after a period of speed endurance training. It could be speculated that the elevated level of α_2 -isoforms in the SET group after the IT period may have increased the number of functional pumps, causing a lower accumulation of K^+ in the muscle interstitium during exercise and in the recovery from exercise (47, 48). In support of this, the plasma K^+ concentration was lower at the end of the repeated intense exercise bouts after the IT period. In agreement, in the study of Iaia et al. (31), the elevated level of α_1 -subunits after a period with sprint training was associated with a lower rate of increase in venous K^+ concentration during an intense exercise bout, and the lowering of plasma K^+ concentration 1 min after the intense exercise was correlated with the level of muscle Na^+-K^+ pump α_1 -subunits (31). It may be that the amount of α -subunits are limiting for the formation of Na^+-K^+ pumps and that the elevated level of α_1 -subunits in the study of Iaia et al. (31) and α_2 -subunits in the present study after the IT period have lead to an increased

content of active Na^+-K^+ pumps during exercise. Consequently, the reduction in muscle membrane potential may have been lowered and cell excitability preserved (10), and thereby the time to fatigue during supramaximal exercise is prolonged. Together, these findings support a role of muscle Na^+-K^+ pumps in the control of extracellular K^+ concentration and fatigue development during intense exercise. The expression of NKCC1 was not changed in the present study. Increases of 14–29% have been observed when training rats (22), and Iaia et al. (31) found in humans a nonsignificant 14% increase in NKCC1 with a reduced volume of training and sprint training. Apparently, a further reduction in training may have been needed for the amount of NKCC1 change, and alterations in NKCC1 expression cannot explain the slower development of muscle fatigue during intense exercise after the IT period.

The expression of muscle NHE1 was not changed in the present study, which is in contrast to the finding of Iaia et al. (31) and studies where untrained subjects performed a period of sprint training (34, 45). Apparently, the greater volume of training in the present study impaired the net synthesis of NHE1. Similarly, neither MCT1 nor MCT4 were changed with training, which is in accordance with the observations of Iaia et al. (31). On the other hand, most other studies (5, 8, 34, 45, 52) with untrained subjects using high-intensity intermittent training have shown a higher amount of MCT1, and one study (5) has reported sprint training-induced changes in MCT1 proteins in endurance-trained subjects. In that study, the subjects maintained a high volume of training (~50 km/wk). It may be that the subjects in the present study already had an elevated MCT1 protein content before the change in training since endurance training has been shown to increase MCT1 density (7, 14, 23) and lactate transport capacity has been observed to be higher in trained compared with untrained subjects (51). The finding of unaltered MCT4 levels is consistent with the majority of the other studies (5, 16, 34, 45). The unaltered NHE1 and MCT protein expression observed after the IT period is consistent with the finding of unchanged blood lactate concentrations during the SUB and RS tests. The finding of higher blood lactate levels in the recovery period

from the INC test is probably a reflection of the more work performed after the IT period. Apparently, improved short-term performance can occur without changes in some of the key H^+ transport proteins.

In summary, the present study showed that speed endurance training together with a reduction in total volume of training lead to both improved long- and short-term performance, which was associated with a significant elevated level of Na^+-K^+ pump α_2 -subunits and lower plasma K^+ concentrations during exhaustive running.

Perspectives and Significance

The present study examined the muscular effects and performance aspects in relation to a change from regular endurance training to a reduced amount of training and additional speed endurance training. Significant changes in both long- and short-term performance were observed, which was associated with an increase in the Na^+-K^+ pump α_2 -subunit, whereas a number of metabolic enzymes and other key muscle ion transport proteins were unaltered. Future research should try to elucidate what are causing the changes and study the importance of muscle ion regulation for work capacity during exercise at various intensities. This study has important practical implications, as it suggests that in already trained subjects, further muscle adaptations can occur and performance can be improved by adding speed endurance training. Endurance runners may, therefore, in some periods, benefit from replacing overall volume of training with sessions of high-intensity exercise. This information is clearly of great interest not only for elite athletes but also for people participating in recreational activities. Specifically, in a health perspective, a prospective study (1) on United States male physicians has suggested that habitual vigorous exercise, as in the present study, diminishes the risk of sudden death during vigorous exertion.

ACKNOWLEDGMENTS

The authors thank Jens Jung Nielsen and Dorte Jessing Agerby Hanskov for excellent technical assistance and Henning Bundgaard for providing the McB2 Na^+-K^+ pump α_2 -subunit antibody.

GRANTS

This work was supported by the Danish Natural Science Research Council Grant 272-05-0407, Team Denmark, and the Ministry of Culture (Kulturministeriets Udvalg for Idrætsforskning).

DISCLOSURES

No conflicts of interest are declared by the author(s).

REFERENCES

- Albert CM, Mittleman MA, Chae CU, Lee IM, Hennekens CH, Manson JE. Triggering of sudden death from cardiac causes by vigorous exertion. *N Engl J Med* 343: 1355–1361, 2000.
- Aughey RJ, Murphy KT, Clark SA, Garnham AP, Snow RJ, Cameron-Smith D, Hawley JA, McKenna MJ. Muscle Na^+-K^+ -ATPase activity and isoform adaptations to intense interval exercise and training in well-trained athletes. *J Appl Physiol* 103: 39–47, 2007.
- Bangsbo J, Juel C. Counterpoint: lactic acid accumulation is a disadvantage during muscle activity. *J Appl Physiol* 100: 1412–1413, 2006.
- Bergstrom J. Muscle electrolytes in man. *Scand J Clin Lab Invest* 68: 1–110, 1962.
- Bickham DC, Bentley DJ, Le Rossignol PF, Cameron-Smith D. The effects of short-term sprint training on MCT expression in moderately endurance-trained runners. *Eur J Appl Physiol* 96: 636–643, 2006.
- Billat VL, Flechet B, Petit B, Muriaux G, Koralsztein JP. Interval training at $\dot{V}O_{2max}$: effects on aerobic performance and overtraining markers. *Med Sci Sports Exerc* 31: 156–163, 1999.
- Bonon A, McCullagh KJ, Putman CT, Hultman E, Jones NL, Heigenhauser GJ. Short-term training increases human muscle MCT1 and femoral venous lactate in relation to muscle lactate. *Am J Physiol Endocrinol Metab* 274: E102–E107, 1998.
- Burgomaster KA, Cermak NM, Phillips SM, Benton CR, Bonon A, Gibala MJ. Divergent response of metabolite transport proteins in human skeletal muscle after sprint interval training and detraining. *Am J Physiol Regul Integr Comp Physiol* 292: R1970–R1976, 2007.
- Burgomaster KA, Hughes SC, Heigenhauser GJ, Bradwell SN, Gibala MJ. Six sessions of sprint interval training increases muscle oxidative potential and cycle endurance capacity in humans. *J Appl Physiol* 98: 1985–1990, 2005.
- Cairns SP, Lindinger MI. Do multiple ionic interactions contribute to skeletal muscle fatigue? *J Physiol* 586: 4039–4054, 2008.
- Clausen T. Na^+-K^+ pump regulation and skeletal muscle contractility. *Physiol Rev* 83: 1269–1324, 2003.
- Dela F, Holten M, Juel C. Effect of resistance training on Na,K pump and Na^+/H^+ exchange protein densities in muscle from control and patients with type 2 diabetes. *Pflugers Arch* 447: 928–933, 2004.
- Di Prampero PE. The energy cost of human locomotion on land and in water. *Int J Sports Med* 7: 55–72, 1986.
- Dubouchaud H, Butterfield GE, Wolfel EE, Bergman BC, Brooks GA. Endurance training, expression, and physiology of LDH, MCT1, and MCT4 in human skeletal muscle. *Am J Physiol Endocrinol Metab* 278: E571–E579, 2000.
- Esfarjani F, Laursen PB. Manipulating high-intensity interval training: effects on $\dot{V}O_{2max}$, the lactate threshold and 3000 m running performance in moderately trained males. *J Sci Med Sport* 10: 27–35, 2007.
- Evertsen F, Medbo JJ, Bonon A. Effect of training intensity on muscle lactate transporters and lactate threshold of cross-country skiers. *Acta Physiol Scand* 173: 195–205, 2001.
- Evertsen F, Medbo JJ, Jebens E, Nicolaysen K. Hard training for 5 mo increases Na^+-K^+ pump concentration in skeletal muscle of cross-country skiers. *Am J Physiol Regul Integr Comp Physiol* 272: R1417–R1424, 1997.
- Fitts RH. The muscular system: fatigue processes. In: *ACSM's Advanced Exercise Physiology*, edited by Tipton CM. Philadelphia, PA: Lippincott, Williams & Wilkins, 2006, p. 178–196.
- Fowles JR, Green HJ, Ouyang J. Na^+-K^+ -ATPase in rat skeletal muscle: content, isoform, and activity characteristics. *J Appl Physiol* 96: 316–326, 2004.
- Foxdal P, Bergqvist Y, Eckerbom S, Sandhagen B. Improving lactate analysis with the YSI 2300 GL: hemolyzing blood samples makes results comparable with those for deproteinized whole blood. *Clin Chem* 38: 2110–2114, 1992.
- Franch J, Madsen K, Djurhuus MS, Pedersen PK. Improved running economy following intensified training correlates with reduced ventilatory demands. *Med Sci Sports Exerc* 30: 1250–1256, 1998.
- Gosmanov AR, Nordtvedt NC, Brown R, Thomason DB. Exercise effects on muscle beta-adrenergic signaling for MAPK-dependent NKCC activity are rapid and persistent. *J Appl Physiol* 93: 1457–1465, 2002.
- Green H, Halestrap A, Mockett C, O'Toole D, Grant S, Ouyang J. Increases in muscle MCT are associated with reductions in muscle lactate after a single exercise session in humans. *Am J Physiol Endocrinol Metab* 282: E154–E160, 2002.
- Green HJ, Barr DJ, Fowles JR, Sandiford SD, Ouyang J. Malleability of human skeletal muscle Na^+-K^+ -ATPase pump with short-term training. *J Appl Physiol* 97: 143–148, 2004.
- Green HJ, Chin ER, Ball-Burnett M, Ranney D. Increases in human skeletal muscle Na^+-K^+ -ATPase concentration with short-term training. *Am J Physiol Cell Physiol* 264: C1538–C1541, 1993.
- He S, Shelly DA, Moseley AE, James PF, James JH, Paul RJ, Lingrel JB. The α_1 - and α_2 -isoforms of Na,K -ATPase play different roles in skeletal muscle contractility. *Am J Physiol Regul Integr Comp Physiol* 281: R917–R925, 2001.
- Helgerud J, Hoydal K, Wang E, Karlsen T, Berg P, Bjerkaas M, Simonsen T, Helgesen C, Hjorth N, Bach R, Hoff J. Aerobic high-intensity intervals improve $\dot{V}O_{2max}$ more than moderate training. *Med Sci Sports Exerc* 39: 665–671, 2007.

28. **Houston ME, Thomson JA.** The response of endurance-adapted adults to intense anaerobic training. *Eur J Appl Physiol Occup Physiol* 36: 207–213, 1977.
29. **Hundal HS, Maxwell DL, Ahmed A, Darakhshan F, Mitsumoto Y, Klip A.** Subcellular distribution and immunocytochemical localization of Na,K-ATPase subunit isoforms in human skeletal muscle. *Mol Membr Biol* 11: 255–262, 1994.
30. **Iaia FM, Hellsten Y, Nielsen JJ, Fernstrom M, Sahlin K, Bangsbo J.** Four weeks of speed endurance training reduces energy expenditure during exercise and maintains muscle oxidative capacity despite a reduction in training volume. *J Appl Physiol* 106: 73–80, 2009.
31. **Iaia FM, Thomassen M, Kolding H, Gunnarsson T, Wendell J, Rostgaard T, Nordsborg N, Krstrup P, Nybo L, Hellsten Y, Bangsbo J.** Reduced volume but increased training intensity elevates muscle Na⁺-K⁺ pump α 1-subunit and NHE1 expression as well as short-term work capacity in humans. *Am J Physiol Regul Integr Comp Physiol* 294: R966–R974, 2008.
32. **Juel C.** Lactate-proton cotransport in skeletal muscle. *Physiol Rev* 77: 321–358, 1997.
33. **Juel C.** Training-induced changes in membrane transport proteins of human skeletal muscle. *Eur J Appl Physiol* 96: 627–635, 2006.
34. **Juel C, Klarskov C, Nielsen JJ, Krstrup P, Mohr M, Bangsbo J.** Effect of high-intensity intermittent training on lactate and H⁺ release from human skeletal muscle. *Am J Physiol Endocrinol Metab* 286: E245–E251, 2004.
35. **Kjeldsen K, Richter EA, Galbo H, Lortie G, Clausen T.** Training increases the concentration of [³H]ouabain-binding sites in rat skeletal muscle. *Biochim Biophys Acta* 860: 708–712, 1986.
36. **Klitgaard H, Clausen T.** Increased total concentration of Na-K pumps in vastus lateralis muscle of old trained human subjects. *J Appl Physiol* 67: 2491–2494, 1989.
37. **Ladka R, Ng YC.** Na⁺-transport modulation induces isoform-specific expression of Na⁺,K⁺-ATPase α -subunit isoforms in C2C12 skeletal muscle cell. *Mol Cell Biochem* 211: 79–84, 2000.
38. **Laursen PB, Jenkins DG.** The scientific basis for high-intensity interval training: optimising training programmes and maximising performance in highly trained endurance athletes. *Sports Med* 32: 53–73, 2002.
39. **Lowry OH, Passonneau JV.** *A Flexible System of Enzymatic Analysis*. New York: Academic, 1972, p. 237–249.
40. **Madsen K, Franch J, Clausen T.** Effects of intensified endurance training on the concentration of Na,K-ATPase and Ca-ATPase in human skeletal muscle. *Acta Physiol Scand* 150: 251–258, 1994.
41. **McKenna MJ.** Effects of training on potassium homeostasis during exercise. *J Mol Cell Cardiol* 27: 941–949, 1995.
42. **Medbo JJ, Jebens E, Vikne H, Refsnes PE, Gramvik P.** Effect of strenuous strength training on the Na-K pump concentration in skeletal muscle of well-trained men. *Eur J Appl Physiol* 84: 148–154, 2001.
43. **Mikkola J, Rusko H, Nummela A, Pollari T, Hakkinen K.** Concurrent endurance and explosive type strength training improves neuromuscular and anaerobic characteristics in young distance runners. *Int J Sports Med* 28: 602–611, 2007.
44. **Millet GP, Jaouen B, Borrani F, Candau R.** Effects of concurrent endurance and strength training on running economy and $\dot{V}O_2$ kinetics. *Med Sci Sports Exerc* 34: 1351–1359, 2002.
45. **Mohr M, Krstrup P, Nielsen JJ, Nybo L, Rasmussen MK, Juel C, Bangsbo J.** Effect of two different intense training regimens on skeletal muscle ion transport proteins and fatigue development. *Am J Physiol Regul Integr Comp Physiol* 292: R1594–R1602, 2007.
46. **Muto S, Nemoto J, Okada K, Miyata Y, Kawakami K, Saito T, Asano Y.** Intracellular Na⁺ directly modulates Na⁺,K⁺-ATPase gene expression in normal rat kidney epithelial cells. *Kidney Int* 57: 1617–1635, 2000.
47. **Nielsen JJ, Mohr M, Klarskov C, Kristensen M, Krstrup P, Juel C, Bangsbo J.** Effects of high-intensity intermittent training on potassium kinetics and performance in human skeletal muscle. *J Physiol* 554: 857–870, 2004.
48. **Nordsborg N, Ovesen J, Thomassen M, Zangenberg M, Jons C, Iaia FM, Nielsen JJ, Bangsbo J.** Effect of dexamethasone on skeletal muscle Na⁺,K⁺ pump subunit specific expression and K⁺ homeostasis during exercise in humans. *J Physiol* 586: 1447–1459, 2008.
49. **Nordsborg N, Thomassen M, Lundby C, Pilegaard H, Bangsbo J.** Contraction induced increases in Na⁺-K⁺-ATPase mRNA levels in human skeletal muscle are not amplified by activation of additional muscle mass. *Am J Physiol Regul Integr Comp Physiol* 289: R84–R91, 2005.
50. **Paavolainen L, Hakkinen K, Hamalainen I, Nummela A, Rusko H.** Explosive-strength training improves 5-km running time by improving running economy and muscle power. *J Appl Physiol* 86: 1527–1533, 1999.
51. **Pilegaard H, Bangsbo J, Richter EA, Juel C.** Lactate transport studied in sarcolemmal giant vesicles from human muscle biopsies: relation to training status. *J Appl Physiol* 77: 1858–1862, 1994.
52. **Pilegaard H, Domino K, Noland T, Juel C, Hellsten Y, Halestrap AP, Bangsbo J.** Effect of high-intensity exercise training on lactate/H⁺ transport capacity in human skeletal muscle. *Am J Physiol Endocrinol Metab* 276: E255–E261, 1999.
53. **Porszasz J, Barstow TJ, Wasserman K.** Evaluation of a symmetrically disposed Pitot tube flowmeter for measuring gas flow during exercise. *J Appl Physiol* 77: 2659–2665, 1994.
54. **Saunders PU, Telford RD, Pyne DB, Peltola EM, Cunningham RB, Gore CJ, Hawley JA.** Short-term plyometric training improves running economy in highly trained middle and long distance runners. *J Strength Cond Res* 20: 947–954, 2006.
55. **Sjogaard G.** Water and electrolyte fluxes during exercise and their relation to muscle fatigue. *Acta Physiol Scand Suppl* 556: 129–136, 1986.
56. **Slawinski J, Demarle A, Koralsztein JP, Billat V.** Effect of supra-lactate threshold training on the relationship between mechanical stride descriptors and aerobic energy cost in trained runners. *Arch Physiol Biochem* 109: 110–116, 2001.
57. **Spurrs RW, Murphy AJ, Watsford ML.** The effect of plyometric training on distance running performance. *Eur J Appl Physiol* 89: 1–7, 2003.
58. **Svedenhag J, Sjodin B.** Physiological characteristics of elite male runners in and off-season. *Can J Appl Sport Sci* 10: 127–133, 1985.
59. **Thorstensson A, Sjodin B, Karlsson J.** Enzyme activities and muscle strength after “sprint training” in man. *Acta Physiol Scand* 94: 313–318, 1975.
60. **Turner AM, Owings M, Schwane JA.** Improvement in running economy after 6 weeks of plyometric training. *J Strength Cond Res* 17: 60–67, 2003.
61. **Wolitzky BA, Fambrough DM.** Regulation of the (Na⁺ + K⁺)-ATPase in cultured chick skeletal muscle. Modulation of expression by the demand for ion transport. *J Biol Chem* 261: 9990–9999, 1986.
62. **Wong JA, Gosmanov AR, Schneider EG, Thomason DB.** Insulin-independent, MAPK-dependent stimulation of NKCC activity in skeletal muscle. *Am J Physiol Regul Integr Comp Physiol* 281: R561–R571, 2001.